

Multidimensional Trust Evaluation and Task Match based Workers Recruitment Scheme for MCS

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Abstract—Recruiting trust workers to achieve high data quality at low cost has become a promising approach in Mobile Crowdsensing (MCS). However, most existing trust evaluation methods only adopt a single-dimensional trust model, neglecting the fact that a worker's trustworthiness can vary across different task types, which leads to suboptimal task-worker matching, poor data quality, and inefficient cost utilization. To this end, we propose a Multidimensional Trust Evaluation and Task Matching (MTE-TM) based workers recruitment scheme to improve data quality while reducing costs for MCS. First, we represent worker trustworthiness by a composite of expected trust values and variance, enabling task-specific trust assessment that extends traditional single-dimensional trust to a multidimensional domain. A novel Expectation-Maximization (EM)-based trust evaluation mechanism is also introduced to improve accuracy. Second, we design a new worker selection method that combines a worker's trust level and the width of the confidence interval to compute their Upper Confidence Bound (UCB) index, which effectively guides worker selection toward optimal outcomes. Third, we propose an Optimized Data Quality Matching (ODQM) algorithm that assigns tasks to workers with high priority and low bid prices under budget constraints, thereby further improving data quality. The experimental results demonstrate the significant performance improvements of our scheme, achieving 45.54~95.55% optimization in trust evaluation, 65.01~72.95% improvement in data quality, and a notable reduction in regret.

Index Terms—Mobile crowdsensing; Data quality; Reputation evaluation; Worker selection; Task assignment.

I. INTRODUCTION

MOBILE Crowdsensing (MCS) has emerged as a pivotal paradigm that leverages ubiquitous mobile devices to collect large-scale sensing data, thereby enabling diverse services [1]–[3]. The motivation for adopting MCS stems from its unique advantages over traditional sensing infrastructures [4]. By leveraging the pervasive sensing, computation, and communication capabilities of mobile devices [5], MCS enables large-scale, fine-grained, and real-time data collection with

lower deployment and maintenance costs [6]. Additionally, the inherent spatial-temporal diversity of mobile participants provides richer coverage than static sensor networks. These characteristics make MCS particularly valuable in real-world applications, including intelligent transportation systems, personalized healthcare monitoring [7], public safety [8], emergency response, smart city management, and environmental quality assessment [9].

Generally, data quality is a critical concern for MCS [10]–[12], since data quality directly impacts the reliability of the service [13], [14]. The growing need for cost-effective and high-resolution sensing services further highlights the importance of recruiting trustworthy workers and ensuring high-quality data collection [14], [15]. Trust-based worker recruitment has become a widely adopted approach for achieving high-quality data collection [16]–[18]. Such studies primarily focus on three key issues: (1) Trust evaluation methods, including how worker trustworthiness is measured and represented [19], [20]; (2) Recruitment strategies, which aim at selecting trustworthy workers to report high-quality data [21]; (3) Optimization of task assignment, where optimal data quality and cost efficiency are achieved only when the abilities of the assigned workers match the task requirements precisely [22], [23]. However, existing trust-based data collection methods still face significant limitations in these three aspects.

First, current methods fail to comprehensively reflect the trustworthiness of workers, leading to suboptimal data quality. In practical MCS scenarios, workers' sensing capabilities vary due to heterogeneous devices [24] and domain-specific expertise [25], [26], resulting in task-dependent trust levels. Current studies oversimplify this by representing trust as a single-dimensional metric, which cannot reflect workers' actual performance across diverse tasks.

Second, existing trust-based worker selection methods fail to guarantee high-quality worker recruitment. To illustrate this, we demonstrate how current single-dimensional trust evaluation approaches tend to select mediocre performers across all domains while excluding workers who excel in specific domains but perform moderately in others. This ultimately leads to suboptimal data quality [25], [26]. Consider workers A, B, C who achieve excellent quality (0.99) in domains $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$, respectively, while maintaining baseline performance (0.88) in other domains. Their single-dimensional trust scores (average: 0.9075) are lower than worker D , who performs uniformly at 0.91 in domains $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$. Under single-dimensional evaluation, D is selected despite inferior domain-specific quality (0.91 vs. 0.99), exemplifying “bad money drives out good.” In contrast, adopting a multidimen-

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sional trust evaluation to represent workers' varying trust levels across different domains can effectively enable precise matching of high-quality workers to tasks. For example, the multidimensional trust levels of worker A would be: (0.99, 0.88, 0.88). When selecting workers for tasks, we can then choose the worker with the highest trust level in that specific domain. Thus, for tasks in domains $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$, the selected workers would be A, B, C , respectively. In this case, the achieved data quality is 0.99, representing the optimized result.

Third, current studies often fail to achieve optimal system performance due to poor task-worker matching [27], [28]. Even under single-dimensional trust evaluation, achieving high data quality at low cost is inherently a multi-objective optimization problem [28]. However, existing approaches typically simplify this by converting it into a single-objective metric, such as the trustworthiness-to-bid ratio [29], which oversimplifies the optimization problem and fails to deliver near-optimal results [30], [31]. In real MCS scenarios, the trustworthiness of a worker is multidimensional [25], [26]. To achieve better optimization, tasks must be assigned to workers who exhibit high domain-specific reliability and low bid prices, which is the key problem addressed in this paper.

To address the above issues, we propose a Multidimensional Trust Evaluation and Task Matching (MTE-TM) workers recruitment scheme to improve data quality and reduce cost. The main contributions of our research are summarized as follows:

(1) **Multidimensional Trust Evaluation Framework:** We propose a novel multidimensional trust model that characterizes worker credibility through both reputation scores and probabilistic trust distributions, enabling precise trustworthiness quantification. Unlike traditional single-dimensional metrics, our framework captures task-specific competence variations across multiple domains. An EM-based trust assessment mechanism iteratively refines worker trust parameters, thereby enhancing evaluation accuracy.

(2) **Trust-aware Worker Selection Scheme:** A multidimensional Upper Confidence Bound (UCB) index is designed to integrate a worker's trustworthiness and confidence interval width. The use of domain-specific indices enables the selection of workers who excel in particular task domains rather than relying solely on overall trust ratings. This design balances exploration and exploitation, reduces repeated selections, and provides a clear mathematical foundation.

(3) **Optimized Data Quality Matching (ODQM) Algorithm:** We propose the ODQM algorithm to efficiently assign tasks to high-priority and low-bid workers, enhancing task execution efficiency and data quality while maintaining budget compliance and meeting workers' reward expectations.

The remainder of this paper is organized as follows. In Section II, we review the existing work. Section III presents the system model and problem formulation. The MTE-TM scheme is proposed in detail in Section IV. Performance analysis is provided in Section V. Finally, we give the conclusion and future work in Section VI.

II. RELATED WORK

MCS has emerged alongside the rise of edge computing [32], serving as an effective paradigm for data collection

and processing through the integration of cloud-edge computing infrastructures [33]. Data quality is a critical factor in MCS, directly affecting the performance of data-driven services [10]–[13]. Low-quality or erroneous data can severely degrade service performance, leading to losses for users. For example, incorrect data may cause navigation services to recommend inefficient routes, resulting in additional time and economic costs. To address this, extensive research has been conducted to improve data quality. In the following, this section reviews relevant studies.

A. Quality-aware data collection scheme

In MCS, worker quality reflects the overall reliability of their reported data, often termed *trustworthiness* [33]. Thus, high-quality data collection requires selecting workers based on their trustworthiness [34], [35]. Since worker quality is initially unknown, platforms must evaluate it by assessing the submitted data [36], which is reasonable. Some studies further propose estimating data quality through the performance of the derived services. However, obtaining high-quality data is not trivial. A key challenge lies in achieving globally optimal worker selection, as the platform typically recruits only a small subset of available workers [31], [34], [36], meaning most workers' quality remains unknown. Consequently, selections based solely on known high-quality workers may lead to suboptimal, locally optimal results rather than achieving global optimality.

To address this, many studies have proposed exploration-exploitation strategies [11], [30], [31], [34], [36]. Exploitation refers to the selection of known high-quality workers, while exploration refers to the random selection of unknown workers to discover potentially trustworthy workers. The most direct approach employing this strategy is the pure greedy algorithm [37]. The pure greedy algorithm assigns a fixed exploration round to each worker and then selects the top performers. However, in addition to the issue of undetermined exploration rounds, it suffers from an inefficiency: some workers with evidently poor quality may not require extensive exploration, yet the algorithm still assigns them fixed trials, leading to unnecessary costs. The epsilon-greedy algorithm is a probability-based balancing strategy [38]. In the epsilon-greedy algorithm, the exploration/exploitation trade-off is determined by a predefined probability ϵ . However, since the epsilon-greedy algorithm operates purely randomly without leveraging historical data, it may unnecessarily explore clearly untrustworthy workers, similar to the pure greedy approach, thus incurring additional costs [38]. Among exploration-exploitation methods, the Multi-Armed Bandit (MAB) framework has gained widespread adoption [11], [30], [31], [34], [36]. The MAB strategy selects workers using the Upper Confidence Bound (UCB) index to determine their selection probabilities [11], [30], [31]. Unlike pure greedy and epsilon-greedy algorithms, the UCB algorithm leverages prior exploration data to quickly discard low-trust workers while still giving underexplored, uncertain workers a chance to stand out. This is achieved through the UCB index, which consists of two main components: worker quality and a function

related to the number of times a worker has been selected. Higher-quality workers have larger UCB values and thus higher selection probabilities. Conversely, the more frequently a worker has been selected, the lower their probability of being chosen again, encouraging the exploration of less-known workers [11], [30], [31], [34], [36].

However, current research has several limitations. First, the core idea behind the UCB index is that the quality of a worker can be estimated with high confidence after being selected multiple times, allowing for reduced future selection. Yet, existing UCB index designs do not directly reflect confidence in the estimated quality. Instead, they use a heuristic function based solely on the number of times a worker has been selected, lacking a solid mathematical foundation. In reality, workers with stable quality (i.e., narrow confidence intervals) require fewer selections to estimate their quality accurately, while those with wider intervals need more. Using the same function for all workers can lead to inaccurate estimates for some and unnecessary selection costs for others. To address this limitation, some studies suggest incorporating the confidence interval directly into the UCB index design, which provides a more rigorous mathematical foundation. Second, current models assign a single trust value to each worker across all domains, which fails to reflect the variability in a worker's performance across different fields and the corresponding differences in exploration requirements. In our approach, workers have domain-specific trust values, and the UCB index is calculated separately for each domain. This enables more accurate selection and improves overall data quality.

Existing multidimensional approaches, such as MLM-WR [25] and CMAB-based truth discovery [30] (referred to as CMBA-HS++), also aim to improve data quality in MCS systems. MLM-WR uses UAVs to obtain ground-truth data, which is costly and impractical in many scenarios. CMAB-based methods assume that the platform knows the post-collection data quality and inject privacy-aware noise, which introduces additional uncertainty. These assumptions limit their practicality in real-world deployment. In contrast, our proposed MTE-TM scheme adopts a multidimensional trust evaluation model capturing both attribute-level and location-level reliability. It dynamically updates trust over multiple rounds and explicitly models worker heterogeneity, enabling effective worker selection and task assignment without relying on UAVs or unrealistic assumptions about post-collection data quality. As summarized in Table I, our approach addresses the limitations of existing methods and provides a more practical and accurate solution.

B. Task-Worker Matching Scheme

Assigning suitable workers to tasks is a challenging issue. In practice, the performance of Mobile Crowdsensing (MCS) depends not only on data quality, but also on multiple optimization objectives, such as cost, coverage, and delay, making it a multi-objective optimization problem [28]. For instance, platforms typically prefer workers who offer high quality at low cost. However, there is typically a trade-off between

TABLE I
COMPARISON WITH EXISTING APPROACHES

Aspect	MLM-WR [25]	CMAB-TD [30]	MTE-TM (Ours)
Trust Modeling	Task-level, UAV GTD	Task-level, noise, post-quality assumed	Multidimensional, attribute & location, dynamic
Ground Truth	UAV-obtained, costly	Assumed post-collection	Not required
Temporal Adaptation	Static	Partial via CMAB	Fully dynamic
Worker Heterogeneity	Implicit	Not explicit	Explicitly modeled
Optimization Goal	High-quality data	Max cumulative reward	Max truthful workers, minimize deviation, optimize cost

quality and cost, as higher-quality workers tend to charge more [11], [30], [31]. A common approach in existing research is to classify workers according to their *quality-to-cost ratio* and select those with the highest ratios. This metric reflects the quality gained per unit cost, benefiting the platform [11], [30], [31]. However, this strategy does not always yield optimal results.

Further studies take worker assignment as a matching problem, where the goal is to assign m workers to n tasks to minimize platform costs while maximizing task performance. This is a classic resource allocation problem in computing. For example, matching tasks to servers: the execution of n tasks requires multi-dimensional computing resources (e.g., CPU, memory, bandwidth) with varying demands, while m servers possess heterogeneous resource capabilities in terms of CPU, memory, and bandwidth [39]. Therefore, optimal performance can be achieved by assigning tasks to well-matched servers. For instance, task t_i requires high computational power but produces little output data. Assigning it to a server with strong computing capability and low bandwidth that matches its requirements can fully leverage overall performance [39]. The same principle applies to assigning workers to tasks. However, a critical limitation persists in these studies: while they acknowledge that tasks have multi-dimensional objectives and workers possess multi-dimensional capabilities, they still employ a single-dimensional trust metric to represent overall task completion quality. This approach fails to account for workers' varying performance across different task domains, consequently leading to suboptimal outcomes.

The strategy proposed by Tian et al. [26] represents one of the studies most closely related to our approach. They observed that workers' task completion quality varies across different locations due to varying levels of familiarity, and therefore advocated considering workers' geographical preferences during task assignment. We have also conducted similar research in Ref. [25]. However, a common limitation is shared in these studies: while focusing on workers' preferences, they fail to systematically integrate these preferences into a comprehensive framework that combines trust evaluation, worker selection, and optimal task-worker matching.

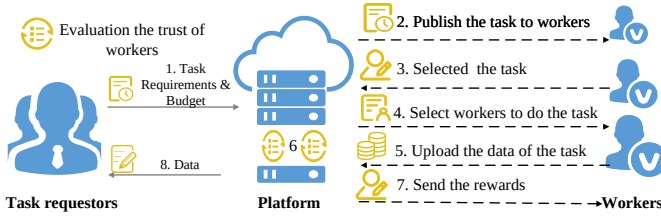


Fig. 1. System model diagram

Although a large body of literature has explored multi-objective optimization in mobile crowdsensing and related fields [40]–[42], the effectiveness of these methods fundamentally depends on whether the underlying problem is accurately characterized in a multidimensional manner [40], [43]. Existing trust-based approaches typically rely on single-dimensional trust metrics, which do not capture workers' heterogeneous competencies, context-dependent behaviors, or location- and attribute-specific reliability [25], [30]. As a result, even if advanced multi-objective optimization techniques are applied, the optimization is performed over an incomplete and low-fidelity representation of worker quality, leading to suboptimal decisions [40]–[42]. In contrast, our work addresses this foundational limitation by first constructing a reliable multidimensional trust evaluation framework that captures workers' performance across multiple task attributes and sensing contexts. This accurate multidimensional characterization provides the necessary foundation for meaningful multi-objective optimization and enables our method to achieve substantially better performance than prior studies relying on single-dimensional trust.

III. SYSTEM MODEL AND DEFINITIONS

A. The System Overview

Without loss of generality, the crowdsensing network model adopted in this paper is shown as Fig. 1. It consists of three main components: Task Requesters (TRs), the Platform, and Workers. The data collection process, shown in Fig. 1, involves eight key steps [16]–[18]: (1) TRs submit tasks and budgets to the platform; (2) The platform publishes tasks to workers; (3) Workers apply for tasks they are willing to perform; (4) The platform selects workers and notifies those chosen to execute the tasks; (5) Workers complete tasks and submit data to the platform; (6) The platform evaluates the quality of the worker (or the trust); (7) The platform compensates workers; (8) The platform delivers data to the TRs. The core issue addressed in this paper is how the platform should evaluate the workers and select the most suitable ones to optimize task completion quality.

Definition 1 (Task, Round). The platform publishes a list of $\mathcal{M}$ tasks $\mathbb{T} = \{T_1, T_2, T_3, \dots, T_{\mathcal{M}}\}$. Task T_i can be represented as a quintuple

$$T_i = \{f_i, Q_i, \tau_i, g_i, B_i\},$$

where f_i represents the domain or type of task (e.g., sensing temperature, humidity, noise). The task types are numbered, so $f_i \in (1, o)$. Q_i represents the minimum expected data quality

requirement for T_i . A higher Q_i demands data closer to the Ground Truth Data (GTD) [19], [21]. τ_i denotes the total number of rounds for T_i . g_i represents the required number of workers for T_i . The calculation is given in Eq. (1):

$$g_i = g e^{-\left(\frac{\mathfrak{P}_k}{\sum_{i=1}^o \mathfrak{P}_i}\right)} \mid k = f_i. \quad (1)$$

Here, e denotes the natural constant, o represents the total number of domains, and $\mathfrak{P}_k$ indicates the historical count of tasks completed by the system for the type of task k ($k = f_i$). $\sum_{i=1}^o \mathfrak{P}_i$ represents the total number of tasks completed in all domains. The parameter g serves as a scaling factor for g_i . From Eq. (1), we observe that when the task completion count for a specific domain is relatively low, the trust evaluation of workers in that domain becomes less certain. Consequently, more workers need to be assigned to such tasks. This approach simultaneously improves the accuracy of task inference and accelerates the speed of trust updating. Additionally, the exponential factor helps mitigate fluctuations in g_i . Specifically, when $\mathfrak{P}_k$ ($k = f_i$) is small, g_i exhibits a high sensitivity to changes. As $\mathfrak{P}_k$ increases, the rate of change in g_i gradually stabilizes, thereby reducing the large fluctuations.

Let B_i denote the task budget allocated for worker payments in T_i . The tasks can be performed in multiple rounds, which are indicated by r .

Definition 2 (Workers). There are $\mathcal{N}$ workers in MCS, $\mathbb{U} = \{u_1, u_2, \dots, u_{\mathcal{N}}\}$. Upon receiving a platform-published task, worker u_j submits a task application represented by a triplet: $\mathbb{A}_{j,i} = \{T_i, b_{j,i}, S_j\}$, where $b_{j,i}$ denotes the expected reward claimed by u_j for executing T_i . S_j represents the number of data units u_j that must be uploaded per round for the current task, and each worker can apply simultaneously for multiple tasks.

B. Problem Formulation

The primary objective of this study is to enable the platform to recruit high-quality workers within specified budget constraints, thereby ensuring high-quality data collection. The foundation for acquiring high-quality workers lies in an accurate trust evaluation. Therefore, we employ two metrics to assess the effectiveness of the strategy in worker evaluation: (1) the accuracy of trust evaluation and (2) the precision of worker classification. These research objectives are quantified through the following four performance indicators:

(1) Data Quality. Data quality is measured by the deviation between the platform-computed results and the ground-truth values, denoted as $\mathfrak{D}_e$.

$$\mathfrak{D}_e = \sum_{i=1}^m \left(\frac{D_i - D_i^*}{D_i^*} \right). \quad (2)$$

Here, m represents the total number of tasks, while D_i and D_i^* denote the computed result and the ground-truth value for task T_i , respectively. The computed result of T_i should approximate the ground truth as closely as possible. Therefore, the optimization objective is given by Eq. (3):

$$\text{Min}(\mathfrak{D}_D) = \text{Min} \left(\frac{\sum_{i=1}^m \frac{|D_i - D_i^*|}{D_i^*}}{m} \right). \quad (3)$$

(2) Budget Saving Rate. While satisfying the task budget constraints, the budget saving rate $\mathfrak{B}$ is defined as follows:

$$\mathfrak{B} = \left(\sum_{i=1}^m \frac{\mathbb{B}_i - \mathbb{B}_i^*}{\mathbb{B}_i} \right) / m. \quad (4)$$

Here, $\mathbb{B}_i$ and $\mathbb{B}_i^*$ denote the allocated budget of T_i and the actual payment to workers upon completion of the task, respectively. For the platform, the payment for T_i must not exceed its allocated budget. Thus, the optimization objective is given by Eq. (5):

$$\text{Min}(\mathfrak{B}) > 0. \quad (5)$$

(3) Accuracy of trust evaluation. The accuracy of worker trust evaluation $\mathfrak{D}_T$ is formulated in Eq. (6):

$$\mathfrak{D}_T = \frac{\sum_{j=1}^n |\mathcal{T}_j - \mathcal{T}_j^*|}{n}. \quad (6)$$

Here, n denotes the total number of workers, and $\mathcal{T}_j$ and $\mathcal{T}_j^*$ represent the estimated and ground-truth trust values of worker u_j , respectively. The evaluated trust values should approximate the true values as closely as possible. Therefore, the optimization objective is expressed in Eq. (7):

$$\text{Min}(\mathfrak{D}_T) = \text{Min} \left(\frac{\sum_{j=1}^n |\mathcal{T}_j - \mathcal{T}_j^*|}{n} \right). \quad (7)$$

(4) Accuracy of the Worker Trust Classification. The classification accuracy is measured by the F_1 -score:

$$\mathcal{F} = \frac{2 \times \mathcal{P} \times \mathcal{R}}{\mathcal{P} + \mathcal{R}}, \quad (8)$$

where $\mathcal{P} = \frac{TP}{TP+FP}$ and $\mathcal{R} = \frac{TP}{TP+FN}$. Here, TP , FP , and FN represent the numbers of true positives, false positives, and false negatives, respectively. The F_1 -score provides a balanced measure of precision and recall. Its maximum value is 1, and the minimum value is 0. A higher score indicates better classification performance. Thus, the optimization objective is given by Eq. (9):

$$\text{Max}(\mathcal{F}) = \text{Max} \left(\frac{2 \times \mathcal{P} \times \mathcal{R}}{\mathcal{P} + \mathcal{R}} \right). \quad (9)$$

In summary, the multi-objective optimization problem is formulated as Eq. (10):

$$\begin{cases} \text{Min}(\mathfrak{D}_D) = \text{Min} \left(\frac{\sum_{i=1}^m |D_i - D_i^*| / D_i^*}{m} \right), \\ \text{Min}(\mathfrak{B}) > 0, \\ \text{Min}(\mathfrak{D}_T) = \text{Min} \left(\frac{\sum_{j=1}^n |\mathcal{T}_j - \mathcal{T}_j^*|}{n} \right), \\ \text{Max}(\mathcal{F}) = \text{Max} \left(\frac{2 \times \mathcal{P} \times \mathcal{R}}{\mathcal{P} + \mathcal{R}} \right). \end{cases} \quad (10)$$

The key notations involved are summarized in Table II.

IV. THE MTE-TM WORKER RECRUITMENT SCHEME

A. Multidimensional Trust Evaluation for Workers

Extending trust evaluation from a single dimension to multiple dimensions introduces several challenges. First, computational complexity increases, as each worker's reliability must be evaluated across multiple task types and locations. Second,

TABLE II
DESCRIPTION OF MAJOR NOTATIONS

Notations	Description
$\mathfrak{D}_T$	Accuracy of trust evaluation
$\mathfrak{D}_{T,c}$	Cycle-normalized trust evaluation accuracy
$\mathcal{F}$	F_1 -score
$\mathcal{F}_c$	Cycle-normalized F_1 -score
$\mathcal{P}_c$	Precision of worker classification in category c
$\mathcal{R}_c$	Recall of worker classification in category c
$\mathfrak{D}_D$	Inference result accuracy
$\mathfrak{D}_{D,c}$	Cycle-normalized inference accuracy
$\mathfrak{D}_{D,c}^*$	Optimal cycle-normalized inference accuracy
$\mathfrak{R}_{D,c}$	Regret per cycle for task allocation
$\mathfrak{B}$	Task budget saving rate
$\mathfrak{B}_c$	Cycle-normalized budget saving rate
$\mathbb{T}$	Set of tasks
m	Total number of tasks
t_i	Task i
$\mathfrak{T}_i$	Tuple describing t_i
$\mathfrak{f}_i$	Domain of t_i
$\mathbb{P}_i$	Data quality requirement of t_i
$\mathfrak{r}_i$	Total rounds for t_i
u_i	Required number of workers for t_i
b_i	Budget of t_i
g_i	Number of workers assigned to t_i
$\mathcal{B}_i$	Budget allocated to t_i
$\mathbb{U}$	Set of workers
n	Total number of workers
u_j	Worker j
$\mathfrak{U}_j$	Tuple describing u_j
$\mathcal{T}_j$	Trust value of u_j
e_j	Expected reward of u_j
$\mathfrak{d}_j$	Data units per round for u_j
$p_{j,k}$	Expected qualification rate of u_j for domain f_k
$I_{j,i}$	Priority of u_j in t_i
r_n	Reputation score of worker n
o	Total number of domains
f_k	Domain k
$\mathcal{C}$	Cycle index / current iteration
$\mathbb{U}_{\mathcal{F}}$	Worker selection strategy under task domains
e	Natural constant

data sparsity in some dimensions can result in high uncertainty in trust estimation. Third, it is necessary to dynamically update trust over multiple rounds to capture variations in worker behavior. Finally, aggregating multidimensional trust scores to guide worker selection and task assignment requires careful design to ensure fairness, stability, and effectiveness. These challenges demonstrate that multidimensional trust evaluation is not a trivial extension from the single-dimensional case and necessitates careful modeling and algorithmic design.

In previous research, worker trustworthiness was measured using a single-dimensional metric based solely on task completion quality. However, there are several limitations of this approach:

(1) A single-dimensional metric fails to reflect the varying

performance of workers across different task types, resulting in suboptimal worker selection. To illustrate this, consider six workers ($u_1, u_2, u_3, u_4, u_5, u_6$) equipped with different sensing devices, whose performance varies significantly across three types of tasks (A, B, C). Table III presents each worker's task-specific quality scores along with their composite scores. Specifically, worker u_5 does not achieve the highest performance in any of the three types of tasks, but consistently ranks second. Consequently, after multiple recruitment rounds, the single-dimensional trust evaluation score of u_5 is 0.87, which is the highest composite quality score among all workers. Worker u_6 shows similarly balanced performance across all task types, although its overall average (0.82) is lower than that of u_5 . In contrast, workers u_1, u_2, u_3, u_4 perform better in 1-2 specialized task types but perform poorly in non-specialized types. This results in their composite scores being lower than both u_5 and u_6 . Thus, in traditional single-dimensional evaluation systems, platforms select workers with the highest composite scores (u_5 and u_6). However, this is not the most optimal result. For example, for Type A tasks, u_1 (quality: 0.98) outperforms u_5 (quality: 0.88); for Type B tasks, u_2 (quality: 0.99) achieves higher quality than u_5 (quality: 0.88); and for Type C tasks, u_4 (quality: 0.99) surpasses u_5 (quality: 0.85). Evidently, evaluating workers with task-type-specific quality metrics, rather than a single-dimensional composite score, allows platforms to select specialists with the highest task-specific performance for each mission type, thereby achieving optimized outcomes.

(2) Single-dimensional trust metrics fail to reflect workers' performance stability, leading to shortcomings in worker selection. Traditional strategies rely solely on average performance to assess workers' abilities. Although simple and intuitive, this approach overlooks a critical stability metric: variance. Average performance reflects overall competence across tasks, but cannot capture performance fluctuations, which are indicative of stability and crucial for predicting future performance. For instance, as shown in Table IV, u_1 has a high average data quality (0.93), but its variance (0.0076) is large, indicating unstable performance, whereas u_2 's average (0.916) is slightly lower, yet its variance is minimal (0.000104), making u_2 preferable for stability-sensitive tasks. The abbreviations used in the last two columns of Table IV are as follows:

- ADQ: Average Data Quality, representing the average performance of a worker across multiple tasks.
- DQV: Data Quality Variance, indicating the variability in the data quality of a worker's tasks.

Therefore, when selecting workers, it is essential to assess not only whether their average performance meets task requirements but also to thoroughly evaluate their performance stability, as measured by variance. Integrating these two dimensions enables more accurate capability assessment and facilitates the identification of workers who can deliver stable and reliable performance for specific tasks. Particularly for missions where data quality stability is critical, prioritizing workers with low variance (high stability) is more important than selecting those solely based on higher average performance.

Furthermore, variations in quality variance among workers

TABLE III
WORKER QUALITY (TRUST) FOR DIFFERENT TASK TYPES AND THE COMPOSITE TRUST

Worker	Type-A tasks	Type-B tasks	Type-C tasks	Composite Trust Score
u_1	0.98	0.67	0.72	0.79
u_2	0.86	0.99	0.63	0.827
u_3	0.63	0.95	0.96	0.847
u_4	0.68	0.83	0.99	0.833
u_5	0.88	0.88	0.85	0.87
u_6	0.82	0.88	0.85	0.85

TABLE IV
THE WORKER'S DATA QUALITY ACROSS DIFFERENT TASKS

Worker	Task-1	Task-2	Task-3	Task-4	Task-5	ADQ	DQV
u_1	1.00	0.85	1.00	0.80	1.00	0.93	0.0076
u_2	0.92	0.90	0.93	0.91	0.92	0.916	0.000104

expose deficiencies in the current UCB index design. The essence of UCB is to estimate the expected quality of a worker through multiple selections, aiming to align the calculated expected quality with the true value. Intuitively, more selections (exploration counts) yield closer approximations. However, increasing exploration incurs higher system costs. Therefore, the goal is to achieve accurate estimates with minimal explorations. The current UCB design uniformly allocates exploration counts to all workers. This approach has two flaws: (a) With few explorations, workers exhibiting high variance obtain inaccurate quality estimates; (b) Excessive explorations for low-variance workers result in unnecessary costs. These issues further demonstrate the inadequacy of single-dimensional trust evaluation metrics.

Therefore, to better characterize workers and enable more rational worker selection, differing from prior studies, MTE-TM represents the trust of worker u_j as a multi-dimensional tuple $\mathcal{T}_j$, as shown in Eq. (11). Here, $\mathcal{T}_{j,k}$ denotes the trust evaluation of worker u_j for the type of task k .

$$\mathcal{T}_j = \{\mathcal{T}_{j,1}, \mathcal{T}_{j,2}, \mathcal{T}_{j,3}, \dots, \mathcal{T}_{j,k}, \dots, \mathcal{T}_{j,o}\}. \quad (11)$$

Moreover, each element $\mathcal{T}_{j,k}$ in the tuple is not a traditional single-dimensional value, but rather represents the distribution of the worker u_j 's data qualification rate. According to the Laplace Central Limit Theorem, the qualification rate of data submitted by worker u_j approximately follows a normal distribution with expectation p and variance $\frac{p(1-p)}{s_j}$. Therefore, the trust evaluation distribution of the worker u_j for the type of task k is given by Eq. (12).

$$\mathcal{T}_{j,k} \sim N\left(\mathcal{P}_{j,k}, \frac{\mathcal{P}_{j,k}(1-\mathcal{P}_{j,k})}{S_{j,k}}\right). \quad (12)$$

Here, $\mathcal{P}_{j,k}$ represents the expected qualification rate of worker u_j for type- k tasks, calculated by Eq. (13):

$$\mathcal{P}_{j,k} = \alpha_{j,k} \times p_{j,k,i} + \beta_{j,k} \times \mathcal{P}_{j,k}'. \quad (13)$$

$$\alpha_{j,k} = \begin{cases} \alpha_{j,k} + \alpha_{\text{step}}, & p_{j,k,i} < \mathcal{P}'_{j,k} \text{ and } \alpha_{j,k} < \alpha_{\max}, \\ \alpha_{j,k}, & p_{j,k,i} = \mathcal{P}'_{j,k}, \\ \alpha_{j,k} - \alpha_{\text{step}}, & p_{j,k,i} > \mathcal{P}'_{j,k} \text{ and } \alpha_{j,k} > \alpha_{\min}. \end{cases} \quad (14)$$

Here, $p_{j,k,i}$ denotes the expected qualification rate of the most recent task data, evaluated based on the data submitted by u_j for the latest type- k task T_i . $\mathcal{P}'_{j,k}$ represents the historically expected qualification rate of u_j for tasks of type- k . The weighting factors $\alpha_{j,k}$ and $\beta_{j,k}$ control the influence of historical data on $\mathcal{P}_{j,k}$, with $\alpha_{j,k} + \beta_{j,k} = 1$. The calculation of $\alpha_{j,k}$ is given by Eq. (14).

Here, the parameter α_{step} denotes the incremental step size used to adjust α , determining how much α increases or decreases during each update. The parameters $\alpha_{\min}$ and $\alpha_{\max}$ represent the lower and upper bounds of α , ensuring that its value remains within a reasonable and controlled range. According to the formula, when the observed performance indicator $p_{j,k,i}$ decreases, the influence of historical data on $\mathcal{P}_{j,k}$ is correspondingly reduced, effectively decreasing the weight of prior observations in updating the worker's performance metric. Conversely, when $p_{j,k,i}$ increases, the influence of historical data is strengthened, allowing past performance to reinforce confidence in current high-quality data. This design balances the update process: it imposes more significant penalties on workers submitting low-quality data while preventing $\mathcal{P}_{j,k}$ from increasing excessively fast over a short period, thereby maintaining stability and robustness in the performance evaluation.

In Eq. (12), $S_{j,k}$ represents the number of data units that worker u_j needs to upload per round for tasks of type k , calculated by Eq. (15):

$$S_{j,k} = e^{-\frac{\mathfrak{P}_{j,k}}{\sum_{i=1}^N \mathfrak{P}_{i,k} \times 10}} \times \mathfrak{d}. \quad (15)$$

Here, $\mathfrak{P}_{j,k}$ represents the number of tasks of type- k already completed by u_j , and $\mathfrak{d}$ is a scaling factor for $S_{j,k}$. As observed from Eq. (15), the value of $S_{j,k}$ depends on the relative number of type- k tasks u_j has completed in the past. When u_j has completed few type- k tasks, trust is relatively uncertain, and thus the value of $S_{j,k}$ increases. This ensures that the variance in the distribution of u_j 's task data qualification rate decreases, meaning the qualification rates across rounds converge closer to the true qualification rate. As a result, the true trust level can be determined more quickly.

Above, a new multi-dimensional worker trust evaluation metric is proposed. However, the specific calculation method for the worker's task qualification rate metric $\mathcal{P}_{j,k}$ has not yet been provided. Therefore, an Expectation-Maximization (EM) based algorithm is proposed to compute $\mathcal{P}_{j,k}$. In this paper, $d_{j,r,x}^i$ denotes the $x|x \in (1..S_{j,k})$ -th data point uploaded by worker $u_j|j \in (1..N)$ in round $r|r \in (1..\tau_i)$ for task T_i . $\mathcal{D}_i$ represents all the data collected for T_i :

$$\mathcal{D}_i = \{d_{j,r,x}^i | j \in (1..N), r \in (1..\tau_i), x \in (1..S_{j,k})\}. \quad (16)$$

Correspondingly, the matrix $\mathcal{D}_i^r$ represents all the data collected for task T_i in round r .

$$\mathcal{D}_i^r = \{d_{j,r,x}^i | r \in (1..\tau_i), x \in (1..S_{j,k})\}. \quad (17)$$

For $d_{j,r,x}^i$, its qualification status is unknown, and the true distribution parameters of u_j 's trust are also unknown. These two issues are interdependent. Once the qualification status of $d_{j,r,x}^i$ is determined, the distribution parameters of u_j 's trust can be estimated using maximum likelihood estimation. Similarly, once the distribution parameters of u_j 's trust are known, the likelihood of $d_{j,r,x}^i$ being qualified can be readily computed.

The EM algorithm is a classical iterative method in machine learning, used to solve maximum likelihood estimation problems involving latent variables. It initializes certain model parameters and iteratively alternates between the Expectation (E) Step and the Maximization (M) Step to update parameter estimates until convergence. In the E-step, the posterior distribution of latent variables is computed, and its expectation is taken based on current parameter estimates. In the M-step, new parameter estimates are obtained by maximizing the log-likelihood function using the expected values of latent variables. Through iterative updates, optimal parameter values are obtained for model application. In this context, $d_{j,r,d}$ is treated as the latent variable $z_{j,r,d}$, and the trust distribution parameters of u_j form the parametric model θ_j . Let $\mathcal{D}_i^r$ denote the dataset collected for task T_i in round r . Let $\mathfrak{f}_i$ denote the task type of T_i . When worker u_j completes a task, their trust evaluation metric for task type $\mathfrak{f}_i$ must be computed and updated.

Before starting the EM algorithm, the trust distribution parameters for each worker on tasks of type $k = \mathfrak{f}_i$ are initialized as $\mu_{j,k} = \mathcal{P}'_{j,k}$ and $\sigma_{j,k} = \sqrt{\frac{\mu_{j,k}(1-\mu_{j,k})}{d_{j,k}}}$, where $\mu_{j,k}$ and $\sigma_{j,k}^2$ represent the expectation and variance of the current trust distribution parameters $\theta_{j,k}$ for worker u_j on tasks of type k .

E-step: In the E-step, the values of latent variables are estimated based on $\mathcal{D}_i^r$ and the current parameter model $\theta_{j,k}$. Specifically, the result of the task per-round $\mathcal{G}_{i,r}$ is estimated by computing a weighted average of the data uploaded by workers, as given by Eq. (18):

$$\mathcal{G}_{i,r} = \frac{\sum_{j=1}^{g_i} \mu_{j,k} \times \sum_{x=1}^{S_{j,k}} d_{j,r,x}^i}{\sum_{j=1}^{g_i} \mu_{j,k} \times S_{j,k}}. \quad (18)$$

Then, using the estimated task results, the qualification of each data point in $\mathcal{D}_i^r$ is determined according to Eq. (19):

$$z_{j,r,x}^i = \begin{cases} 1, & (\mathcal{G}_{i,r} - \varepsilon) \leq d_{j,r,x}^i \leq (\mathcal{G}_{i,r} + \varepsilon), \\ 0, & d_{j,r,x}^i < (\mathcal{G}_{i,r} - \varepsilon) \text{ or } d_{j,r,x}^i > (\mathcal{G}_{i,r} + \varepsilon). \end{cases} \quad (19)$$

Here, $z_{j,r,x}^i$ equals 1 if $d_{j,r,x}^i$ is qualified and 0 if it is unqualified. ε denotes the permissible error margin for the task. Finally, the qualification rate $\mathcal{P}_{j,k}$ for each worker's submitted data in each round is calculated as:

$$\mathcal{P}_{j,k,r} = \frac{\sum_{x=1}^{S_{j,k}} z_{j,r,x}^i}{S_{j,k}}. \quad (20)$$

M-step: In the M-step, we construct and maximize a likelihood function to estimate the trust distribution parameters for each worker. As mentioned previously, the qualification rate $\mathcal{P}_{j,k,r}$ of the data reported by u_j in rounds follows an approximate normal distribution with expectation $\mu_{j,k}$ and variance $\sigma_{j,k}^2$. Thus, its probability density function $f(\mathcal{P}_{j,k,r}; \mu_{j,k}, \sigma_{j,k}^2)$ is given by Eq. (21):

$$f(\mathcal{P}_{j,k,r}; \mu_{j,k}, \sigma_{j,k}^2) = \frac{1}{\sqrt{2\pi} \sigma_{j,k}} \exp\left(-\frac{(\mathcal{P}_{j,k,r} - \mu_{j,k})^2}{2\sigma_{j,k}^2}\right). \quad (21)$$

Where $\sigma_{j,k} = \sqrt{\frac{\mu_{j,k}(1-\mu_{j,k})}{s_{j,k}}}$. For worker u_j in the current round τ_c ($\tau_c \in (1..\tau_i)$), $\mathcal{P}_{j,k,r}$ exists in τ_c . Therefore, the likelihood function $L(\theta_j)$ is constructed as in Eq. (22).

$$\begin{aligned} L(\theta_j) &= \prod_{r=1}^{\tau_c} f(\mathcal{P}_{j,k,r}; \mu_{j,k}, \sigma_{j,k}^2) \\ &= \prod_{r=1}^{\tau_c} \frac{1}{\sqrt{2\pi} \sigma_{j,k}} \exp\left(-\frac{(\mathcal{P}_{j,k,r} - \mu_{j,k})^2}{2\sigma_{j,k}^2}\right) \\ &= (2\pi \sigma_{j,k}^2)^{-\frac{n}{2}} \exp\left(-\sum_{r=1}^{\tau_c} \frac{(\mathcal{P}_{j,k,r} - \mu_{j,k})^2}{2\sigma_{j,k}^2}\right). \end{aligned} \quad (22)$$

Taking the logarithm of both sides of Eq. (22) yields the log-likelihood function $\ln L(\theta_j)$ as in Eq. (23):

$$\ln L(\theta_j) = -\frac{n}{2} \ln 2\pi \sigma_{j,k}^2 - \frac{1}{2\sigma_{j,k}^2} \sum_{r=1}^{\tau_c} (\mathcal{P}_{j,k,r} - \mu_{j,k})^2. \quad (23)$$

Maximizing the log-likelihood function provides the updated trust distribution parameters θ_j for worker u_j , as given by Eq. (24):

$$\begin{aligned} \theta_j &:= \arg \max_{\theta} \ln L(\theta_j) \\ &:= \arg \max_{\theta} \left(-\frac{n}{2} \ln 2\pi \sigma_{j,k}^2 - \frac{1}{2\sigma_{j,k}^2} \sum_{r=1}^{\tau_c} (\mathcal{P}_{j,k,r} - \mu_{j,k})^2 \right). \end{aligned} \quad (24)$$

By iterative execution of the E and M-steps until the convergence condition is satisfied, the expected value of the trust distribution parameter $\mu_{j,k}$ becomes the final estimated value of $\mathcal{P}_{j,k,r}$. At this point, $\mathcal{G}_{i,r}$ represents the inferred result of task T_i in round r .

The worker trust evaluation process is summarized in Algorithm 1. Once worker u_j completes task T_i and updates $\mathfrak{T}_{j,k}$ using Algorithm 1, the worker obtains the latest trust metric for tasks of type k . Consequently, when selecting workers for tasks, those with higher trust in the relevant task category should be prioritized to ensure high-quality data. Furthermore, since $\mathfrak{T}_{j,k}$ is represented as a trust distribution, workers with lower variance can be preferred when trust levels are similar, resulting in higher assurance of data quality. Such nuanced

Algorithm 1 EM-Based Worker Trust Evaluation Process

Input: $\mathcal{D}_i$

Output: $\mathcal{T}_j, \mathcal{G}_{i,r}$

- 1: Extract $\mathfrak{f}_i, Q_i, \tau_i, \mathfrak{g}_i$ from the attributes of task T_i
- 2: Compute $S_{j,k}$ using Eq. (15)
- 3: Let $\mathbb{U}_F$ be the set of workers who submitted data in round r for T_i
- 4: **for** each $u_j \in \mathbb{U}_F$ **do**
- 5: Initialize trust distribution parameters θ_j :
- $\mu_{j,k} = \mathcal{P}'_{j,k}, \quad \sigma_{j,k} = \sqrt{\frac{\mu_{j,k}(1-\mu_{j,k})}{s_{j,k}}}$
- 6: **end for**
- 7: **while** not converged **do**
- 8: # E-step: Compute expected values of latent variables
- 9: Compute the round inference result $\mathcal{G}_{i,r}$ using Eq. (18)
- 10: **for** each data point $d_{j,r,x}^i \in \mathcal{D}_i$ **do**
- 11: Compute $z_{j,r,x}$ via Eq. (19)
- 12: **end for**
- 13: # M-step: Update trust distribution parameters
- 14: **for** each $u_j \in \mathbb{U}_F$ **do**
- 15: Calculate $\mathcal{P}_{j,k,r}$ using Eq. (20)
- 16: Construct the log-likelihood function $\ln L(\theta_j)$ according to Eqs. (21)–(23)
- 17: Update $\theta_{j,k}$ with Eq. (24)
- 18: **end for**
- 19: **end while**
- 20: **for** each $u_j \in \mathbb{U}_F$ **do**
- 21: Assign the converged expected trust value to the current round: $\mathcal{P}_{j,k,r} = \mu_{j,k}$
- 22: Set $\mathcal{P}_{j,k,i} = \mathcal{P}_{j,k,r}$ for task T_i
- 23: Update $\mathfrak{T}_{j,k}$ and $\mathcal{T}_j$ using Eqs. (12)–(14)
- 24: **end for**
- 25: **return** $\mathcal{T}_j, \mathcal{G}_{i,r}$

selection cannot be achieved using previous approaches that rely on single-dimensional trust metrics.

B. The design of Upper Confidence Bound (UCB) index

In worker selection, since the platform only has knowledge of a subset of workers' quality, selecting high-quality workers solely from this known subset yields suboptimal results. To address this, researchers widely adopt an exploration-exploitation approach. Exploitation involves selecting a larger proportion of high-quality workers from the well-known-quality subset, while exploration reserves a smaller proportion for workers from the unknown-quality subset.

During exploration, if high-quality workers are identified, the knowledge of the platform expands, thus increasing the pool of selectable high-quality workers and improving the overall quality of the data. This trade-off is implemented through a UCB index-based selection probability design, in which workers with higher estimated quality receive higher selection probabilities. The selection probability decays with the number of prior selections, ensuring that workers with fewer trials maintain non-zero selection chances.

However, in previous studies, the design of the UCB index employed a single-dimensional trust (quality) to determine the probability $\hat{p}$ of a worker being selected. The calculation formula is generally: $\hat{p} = q_r + \varepsilon_r$, where q_r represents the average quality (i.e., trust level) of the worker in the last r rounds, and ε_r is a function related to the number of times the worker has been selected. However, there are two shortcomings in previous research: (1) The average quality only reflects the historical performance of a worker on tasks and does not guarantee that their completion of the current or next task will necessarily meet this average level, so uncertainty still exists. (2) There is only a single-dimensional overall evaluation metric for all tasks, leading to selecting workers who perform well on average across all tasks rather than excel in specific types of tasks, resulting in suboptimal worker selection.

Therefore, in this paper, a novel multi-dimensional UCB index design method is proposed. (1) Workers' trust is represented as a trust distribution. This distribution can be used to estimate the probability that any worker u_j will successfully complete the current task T_i . This probability, combined with the width of the confidence interval, forms the UCB value $\hat{U}_{j,i}$ for each worker, which determines the priority of worker u_j in completing T_i . (2) The UCB index is extended from single-dimensional to multi-dimensional. In this paper, each worker has an UCB index for every type of task, reflecting their priority for different task categories, enabling the selection of workers who achieve the highest quality for specific tasks, thereby maximizing the quality of task completion under given budget constraints.

Based on the above analysis, the proposed UCB index $\hat{U}_{j,i}$ for worker selection is calculated as shown in Eqs. (25)–(27):

$$\hat{U}_{j,i} = \check{P}_{j,\mathbf{f}_i} + \varepsilon_{j,\mathbf{f}_i}. \quad (25)$$

$$\begin{aligned} \check{P}_{j,\mathbf{f}_i} &= P(Q_{j,\mathbf{f}_i} \geq \mathcal{Q}_i) \\ &= \frac{1}{\sqrt{2\pi}\sigma_{j,\mathbf{f}_i}} \int_{\mathcal{Q}_i}^{+\infty} e^{-\frac{(t-\mu_{j,\mathbf{f}_i})^2}{2\sigma_{j,\mathbf{f}_i}^2}} dt \\ &= 1 - \Phi\left(\frac{q_{j,r,x}^i - \mu_{j,\mathbf{f}_i}}{\sigma_{j,\mathbf{f}_i}}\right). \end{aligned} \quad (26)$$

Here, $Q_{j,\mathbf{f}_i}$ denotes the quality of data $d_{j,\mathbf{f}_i}$ submitted by the worker u_j for the task T_i . The data $d_{j,\mathbf{f}_i}$ in round r consists of multiple data points, where the x -th data point is denoted as $d_{j,r,x}^i$ with quality $q_{j,r,x}^i$. $\mu_{j,\mathbf{f}_i} = \mathcal{P}_{j,\mathbf{f}_i}$, and $\sigma_{j,\mathbf{f}_i} = \sqrt{(\mu_{j,\mathbf{f}_i}(1 - \mu_{j,\mathbf{f}_i}))/S_{j,\mathbf{f}_i}}$. Here, $\mathcal{P}_{j,\mathbf{f}_i}$ represents the probability that worker u_j completes T_i successfully, which depends on the current trust level of u_j . The symbol $\mathbf{f}_i$ indicates the task category to which the task belongs.

$$\varepsilon_{j,\mathbf{f}_i} = \sqrt{\frac{2 \ln \mathfrak{P}_{j,\mathbf{f}_i}}{\mathfrak{P}_{j,\mathbf{f}_i}}}. \quad (27)$$

In Eq. (27), $\varepsilon_{j,\mathbf{f}_i}$ represents the width of the confidence interval of u_j , reflecting its relative exploration frequency and current reliability. From Eq. (28), it can be observed that

when $\mathfrak{P}_{j,\mathbf{f}_i}$ is small (indicating insufficient exploration), $\varepsilon_{j,\mathbf{f}_i}$ increases, resulting in a higher $\hat{U}_{j,i}$. This suggests greater uncertainty about u_j 's true trustworthiness, which warrants more exploration opportunities for T_i . In contrast, as $\mathfrak{P}_{j,\mathbf{f}_i}$ grows, u_j 's estimated trust converges to its true value. Here, the influence of $\varepsilon_{j,\mathbf{f}_i}$ on $\hat{U}_{j,i}$ decreases: workers with high current trust $\check{P}_{j,\mathbf{f}_i}$ gain more opportunities to participate in the task T_i , while low-trust workers are quickly excluded, reflecting exploitation.

C. Optimized Data Quality Matching Algorithm

The set of workers is $\mathbb{U}_1 = \{u_1, u_2, u_3, \dots, u_M\}$, where each worker has its own UCB value $\hat{U}_{j,i}$ and a bid for the task. These two elements form the set: $\mathbb{H}_1 = \{(\hat{U}_{j,i}, \mathbf{b}_{j,i}) \mid j \in \{1 \dots M\}, i \in \{1 \dots N\}\}$. Suppose that there are y workers currently selected for the task T_i , forming the set $\mathbb{U}_i = \{u_1, u_2, u_3, \dots, u_y\}$. The UCB values $\hat{U}_{j,i}$ and bids $\mathbf{b}_{j,i}$ of the workers selected for the task T_i form the set $\mathbb{H}_i = \{(\hat{U}_{1,i}, \mathbf{b}_{1,i}), (\hat{U}_{2,i}, \mathbf{b}_{2,i}), (\hat{U}_{3,i}, \mathbf{b}_{3,i}), \dots, (\hat{U}_{y,i}, \mathbf{b}_{y,i})\}$.

The objective of worker-task matching is to select a set of $\mathbf{g}_i$ workers from $\mathbb{U}_i$ for the task T_i within budget $\mathcal{B}_i$, ensuring high data quality. Let the final selected worker set for the task T_i be $\mathbb{U}_i = \{u_1, u_2, u_3, \dots, u_{\mathbf{g}_i}\}$, where $\sum_{j=1}^{\mathbf{g}_i} \mathbf{b}_{j,i} \leq \mathcal{B}_i$. At the initial time, the selected worker set is empty, i.e., $\mathbb{U}_i = \{\}$.

Worker selection is essentially a multi-objective optimization problem, with two primary goals: selecting workers with higher UCB values and lower expected rewards to ensure high task completion quality and cost-effectiveness. However, most existing strategies only use the trust-to-bid ratio when selecting workers, converting it into a single-dimensional objective, which often fails to guarantee optimal results. To achieve high-quality task completion at a low cost within a limited task budget, MTE-TM proposes the Optimized Data Quality Matching (ODQM) algorithm. The ODQM algorithm comprehensively considers the relationships among workers' UCB values, expected rewards, and task budgets to generate the set of workers $\mathbb{U}_i$ selected for the task T_i . The detailed procedure is as follows:

$$\mathcal{F}(j, \mathcal{B}_i, \mathbf{g}_i, \mathbb{U}_i) = \begin{cases} \hat{U}_{1,i}, \mathbb{U}_i \cup u_1, & \text{if } j = 1 \text{ and } \hat{U}_{1,i} \leq \mathcal{B}_i; \\ 0, & \text{if } j = 1 \text{ and } \hat{U}_{1,i} > \mathcal{B}_i; \\ \hat{U}_{1,i} + \hat{U}_{2,i}, \mathbb{U}_i \cup u_1 \cup u_2, & \\ \text{if } j = 2, (\hat{U}_{1,i} + \hat{U}_{2,i}) \leq \mathcal{B}_i, \mathbf{g}_i \geq 2; \\ \mathcal{F}(1, \mathcal{B}_i, \mathbf{g}_i, \mathbb{U}_i), & \text{if } j = 2, (\hat{U}_{1,i} + \hat{U}_{2,i}) > \mathcal{B}_i; \\ \max(\mathcal{F}(j-1, \mathcal{B}_i, \mathbf{g}_i, \mathbb{U}_i), \mathcal{F}(j-1, \mathcal{B}_i - \mathbf{b}_j, \mathbf{g}_i - 1) \\ + \hat{U}_{j,i}), & \mathbb{U}_i \cup u_j, \text{ otherwise.} \end{cases} \quad (28)$$

Here, $\mathcal{F}(j, \mathcal{B}_i, \mathbf{g}_i, \mathbb{U}_i)$ represents the total UCB value of the optimal worker set composed of the first j workers from $\mathbb{U}_i$ for task T_i , subject to budget $\mathcal{B}_i$ and required worker quantity $\mathbf{g}_i$. $\mathbb{U}_i$ represents the set of selected workers, while $\mathbf{b}_j$ and $\hat{U}_{j,i}$ indicate the expected reward of the worker u_j (the j -th worker in $\mathbb{U}_i$) and its UCB value, respectively. When $j = y$, $\mathbb{U}_i$ becomes the final set of selected workers for T_i .

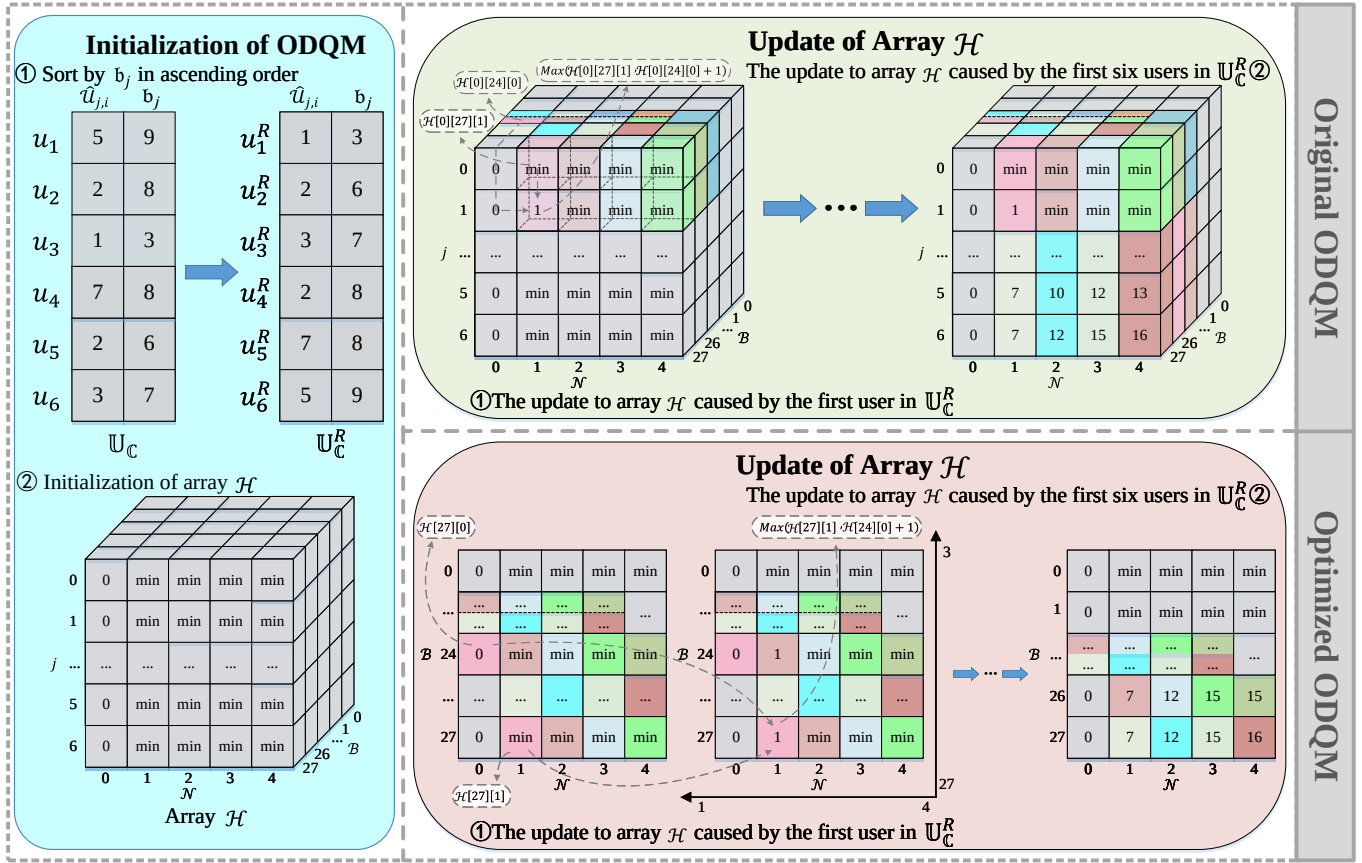


Fig. 2. The process of ODQM

The specific steps of the ODQM algorithm are illustrated in Fig. 2. First, the ODQM algorithm sorts the workers in $\mathcal{U}_C$ in ascending order of their expected rewards, denoted as $\mathcal{U}_C^R = (u_1^R, u_2^R, u_3^R, \dots, u_y^R)$. This step ensures that when the total UCB values are equal, the optimal worker set will have the lowest cost. For clarity, we introduce the array $\mathcal{H}$. Subsequently, the ODQM algorithm initializes the array $\mathcal{H}$ through two main steps: (1) setting all elements in $\mathcal{H}$ to $-\infty$ (representing negative infinity), and (2) setting elements in the array $\mathcal{H}$ to 0 where $g_i = 0$. Finally, the algorithm updates the array $\mathcal{H}$ through three nested loops iterating over: $i_u(1 \rightarrow y)$, $\mathbb{B}(\lfloor b_{j,i} \rfloor \rightarrow \lfloor B_i \rfloor)$, and $\mathcal{N}(1 \rightarrow g_i)$. To address space complexity concerns, the three-dimensional array $\mathcal{H}$ can be reduced to two dimensions for memory optimization. Concurrently, variables $\mathbb{B}$ and $\mathcal{N}$ should be processed by reverse recursion on orders $(\lfloor B_i \rfloor \rightarrow \lfloor b_{j,i} \rfloor)$ and $(g_i \rightarrow 1)$, respectively.

The calculation formula for $\mathcal{H}[\mathbb{B}][\mathcal{N}]$ is given as:

$$\mathcal{H}[\mathbb{B}][\mathcal{N}] = \max(\mathcal{H}_1, \mathcal{H}_2), \quad (29)$$

$$\mathcal{H}_1 = \mathcal{H}[\mathbb{B}][\mathcal{N}], \quad (30)$$

$$\mathcal{H}_2 = \mathcal{H}[\mathbb{B} - b_{j,i}][\mathcal{N} - 1] + \hat{u}_{j,i}. \quad (31)$$

After iterating through all workers in $\mathcal{U}_C^R$, the set of workers $\mathcal{U}[\lfloor B_i \rfloor][g_i]$ used to form element $\mathcal{H}[\mathbb{B}][g_i]$ becomes the final $\mathcal{U}_i$ assigned to T_i . For any worker u_j in $\mathcal{U}_i$, we have:

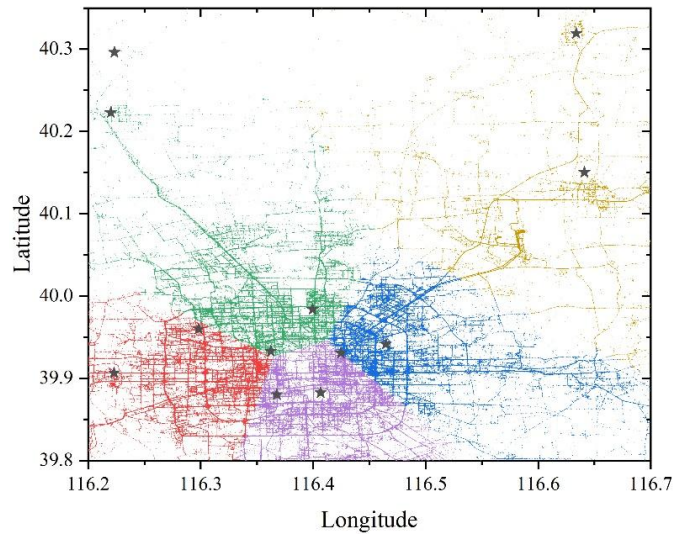


Fig. 3. Trajectory of vehicles in dataset

$$\mathfrak{P}_{j,i} = \begin{cases} \mathfrak{P}_{j,i} + 1, & u_j \in \mathcal{U}_i; \\ \mathfrak{P}_{j,i}, & u_j \notin \mathcal{U}_i. \end{cases} \quad (32)$$

Algorithm 2 demonstrates the detailed procedure for solving $\mathcal{U}_i$ through ODQM.

Algorithm 2 Optimized Data Quality Matching (ODQM) algorithm

Input: $\mathbb{U}_i, \mathbf{f}_i, \mathcal{B}_i, \mathbf{g}_i, \mathbf{b}_{j,i}$
Output: $\mathbb{U}_i, \mathfrak{P}_{j,\mathbf{f}_i}, \mathfrak{P}_{\mathbf{f}_i}$

Initialize: $\mathbb{U}_F = \emptyset, \mathcal{U}_i = \emptyset, \mathbb{H}_i = \emptyset, \mathcal{H}[\cdot] = -\infty, \mathbb{U}[\cdot] = \emptyset$

- 1: **for** each $u_j \in \mathbb{U}_i$ **do**
- 2: Calculate $\hat{\mathcal{U}}_{j,i}$ according to Eqs. (26)–(28)
- 3: $\mathcal{U}_i = \mathcal{U}_i \cup u_j, \mathbb{H}_i = \mathbb{H}_i \cup (\hat{\mathcal{U}}_{j,i}, \mathbf{b}_{j,i})$
- 4: **end for**
- 5: Sort the workers in $\mathcal{U}_i$ in ascending order of expected rewards. Denote the sorted set as $\mathbb{U}_C^R = \{u_1^R, u_2^R, \dots, u_{n_c}^R\}$
- 6: **for** $B = 0$ **to** $\lfloor \mathcal{B}_i \rfloor$ **do**
- 7: **for** $N = 0$ **to** $\mathbf{g}_i$ **do**
- 8: $\mathcal{H}[B][N] = -\infty$
- 9: $\mathbb{U}[B][N] = \emptyset$
- 10: **if** $N = 0$ **then**
- 11: $\mathcal{H}[B][N] = 0$
- 12: **end if**
- 13: **end for**
- 14: **end for**
- 15: **for** $j = 1$ **to** n_c **do**
- 16: **for** $B = \lfloor \mathcal{B}_i \rfloor$ **to** $\lceil \mathbf{b}_{j,i} \rceil$ **step** -1 **do**
- 17: **for** $N = \mathbf{g}_i$ **to** 1 **step** -1 **do**
- 18: $\mathcal{H}[B][N] = \max \left(\mathcal{H}[B][N], \mathcal{H}[B - \mathbf{b}_{j,i}][N - 1] + \hat{\mathcal{U}}_{j,i} \right)$
- 19: Update $\mathbb{U}[B][N] = \mathbb{U}[B - \mathbf{b}_{j,i}][N - 1] \cup u_j$ if max is chosen
- 20: **end for**
- 21: **end for**
- 22: **end for**
- 23: $\mathbb{U}_i = \mathbb{U}[\lfloor \mathcal{B}_i \rfloor][\mathbf{g}_i]$
- 24: **for** each $u_j \in \mathbb{U}_i$ **do**
- 25: Update $\mathfrak{P}_{j,\mathbf{f}_i} = \mathfrak{P}_{j,\mathbf{f}_i} + 1$
- 26: **end for**
- 27: Update $\mathfrak{P}_{\mathbf{f}_i} = \mathfrak{P}_{\mathbf{f}_i} + 1$
- 28: **return** $\mathbb{U}_i, \mathfrak{P}_{j,\mathbf{f}_i}, \mathfrak{P}_{\mathbf{f}_i}$

D. Computational Efficiency Analysis

To rigorously quantify the computational overhead of the proposed MTE-TM framework, we analyze its two dominant modules—EM-based trust inference and the ODQM dynamic programming allocation—and then derive the overall complexity. Let N denote the number of workers, M the number of tasks evaluated per cycle, K the number of trust dimensions, S the upper bound on the number of data samples a worker may submit for a task, I_{EM} the required number of EM iterations, N_{cand} the average number of candidate workers per task after UCB-based filtering, B the discretization granularity of the budget constraint, and g_{max} the maximum number of workers per task.

For EM-based trust inference, each E-step processes all sample-level observations from all participating workers. In the worst case, where every worker contributes at most S samples to every task, each E-step requires $\Theta(MNS)$ compu-

tations, whereas the M-step updates $(\mu_{j,k}, \sigma_{j,k})$ for all workers across all K trust dimensions with $\Theta(NK)$ operations. Since typically $S \gg K/M$, one EM iteration costs $\Theta(MNS)$, and the overall EM complexity is

$$\mathcal{T}_{\text{EM}} = \Theta(I_{\text{EM}} MNS). \quad (33)$$

For ODQM-based allocation, dynamic programming maintains a table of size $B \times g_{\text{max}}$. For each of the N_{cand} candidate workers per task, the algorithm updates states across this table, giving a per-task complexity of $\Theta(N_{\text{cand}} B g_{\text{max}})$. Over M tasks, the total computational cost is

$$\mathcal{T}_{\text{ODQM}} = \Theta(M N_{\text{cand}} B g_{\text{max}}) \approx \Theta(M N_{\text{cand}} B). \quad (34)$$

Thus, the total computational overhead of MTE-TM per evaluation cycle is

$$\mathcal{T}_{\text{total}} = \Theta(I_{\text{EM}} MNS + M N_{\text{cand}} B). \quad (35)$$

In contrast, traditional single-dimensional trust models only require $\Theta(NM)$ operations, indicating that the additional complexity of MTE-TM primarily arises from iterative EM inference and constrained dynamic programming. Nevertheless, this complexity remains practical because: (i) participation sparsity reduces the effective sample size, resulting in $\Theta(I_{\text{EM}} \rho MNS)$ with participation factor ρ ; (ii) UCB filtering ensures $N_{\text{cand}} \ll N$; and (iii) both EM updates and ODQM can be parallelized or computed incrementally. Experimental results further confirm that the runtime is acceptable relative to the substantial gains in trust accuracy and task allocation quality.

V. PERFORMANCE ANALYSIS

A. Experiment Setup

(1) Dataset description. The experiments utilize the Beijing Taxi Trajectory Dataset to simulate the spatial distribution of workers in an MCS network. This dataset contains GPS records from 10,357 taxis operating in Beijing between February 2 and February 8, 2008. Each record includes a taxi ID, timestamp, longitude, and latitude [44]. From this dataset, the trajectories of 2,000 taxis are randomly selected to represent the locations of 2,000 workers. To simulate task ground truth, the Beijing Multi-Site Air-Quality Dataset is used. This dataset contains measurements from 12 air quality monitoring stations in Beijing, collected between March 1, 2013, and February 28, 2017 [45]. Each record comprises six primary air pollutants and six meteorological variables. Eight variables are selected to represent the eight task domains: $PM_{2.5}$ ($\mu\text{g}/\text{m}^3$), PM_{10} ($\mu\text{g}/\text{m}^3$), SO_2 ($\mu\text{g}/\text{m}^3$), NO_2 ($\mu\text{g}/\text{m}^3$), Temperature ($^\circ\text{C}$), Air Pressure (hPa), Dew Point ($^\circ\text{C}$), and Wind Speed (m/s). To model domain-specific participation, the urban area is evenly divided into five regions, and a worker can participate in a task only if they are located within the corresponding region. Fig. 3 illustrates a sample of taxi trajectories over one week, the 12 task locations, and the spatial division of the city. Additional experimental settings are summarized in Table V.

(2) Task and Worker Parameter Settings. (a) The system contains 12 tasks. Each generated task is randomly assigned to a domain, with its minimum required data qualification rate

TABLE V
PARAMETERS FOR EXPERIMENT

Parameter	Value
Total number of domains	8
Total number of urban areas	5
Total number of task locations	12
Total number of trustworthy workers	1200
Total number of unknown workers	600
Total number of malicious workers	200
Expected task data qualification rate of trustworthy workers	0.9–1
Expected task data qualification rate of unknown workers	0.1–0.9
Expected task data qualification rate of malicious workers	0–0.1
Minimum data quality requirement for tasks	50%–80%
Initial trust value of the users	0.1–0.9
Coordination factors of u_j $[\alpha_{j,k}, \beta_{j,k}], [\alpha_j, \beta_j], \alpha_{\text{step}}$	$[0.9, 0.1], 0.1$

randomly generated within $[0.5, 0.8]$. (b) In each domain, 60%, 30%, and 10% of workers are randomly selected as trustworthy, unknown, and malicious workers, respectively. Their expected task data qualification rates are $[0.9, 1]$, $[0.1, 0.9]$, and $[0, 0.1]$. (c) To simulate real-world MCS environments and reflect individual worker differences, workers' initial trust values are randomly generated within $[0, 1]$. (d) The initial values of coordination factors $[\alpha_{j,k}, \beta_{j,k}]$, $[\alpha_j, \beta_j]$, as well as the value of α_{step} , are set to $[0.9, 0.1]$ and 0.1, respectively.

(3) Key Performance Metrics for Experimental Comparison. The performance of the proposed MTE-TM is evaluated using the following metrics: (a) Accuracy of trust evaluation $\mathcal{D}_{\mathcal{T}}$. (b) F_1 score $\mathcal{F}$. (c) Accuracy of inference results $\mathcal{D}_D$. Additionally, to assess MTE-TM's effectiveness in exploration and exploitation, we introduce a regret metric, which represents the difference in inferred task results when using the theoretical optimal worker combination versus MTE-TM's worker combination, denoted as $\mathcal{R}_D$. To evaluate algorithm scalability, we test these metrics across different domain counts, examining the algorithm's ability to handle high-dimensional, complex problems and observing the impact of increasing problem scale on performance.

(4) Baseline Methods for Comparison. To objectively verify and evaluate MTE-TM's performance, we compare it with four different incentive mechanisms:

- **RADP** (Reverse Auction based Dynamic Price): This mechanism does not include trust evaluation. In RADP, workers can sell sensed data to service providers based on their bids. The platform aims to minimize task costs while preserving workers' engagement incentives [46].
- **TRIM** (Task-Oriented User Selection Incentive Mechanism): A single-dimensional trust-based method that converts workers' task completion quality into trust values to identify and exclude low-quality malicious workers, and allocates tasks based on this metric [21], [36].
- **CMBA-HS**: This method uses a traditional UCB-based worker selection strategy. In CMBA-HS, the UCB value of a worker consists of their estimated current data qualification rate and the corresponding confidence interval width [21]. This method considers the historical performance and uncertainty of workers, balancing exploration and exploitation.
- **CMBA-HS++**: An improved version of CMBA-HS that

TABLE VI
OPTIMIZATION DEGREE OF MTE-TM SCHEME ON $\mathcal{D}_{T,c}$ COMPARED WITH OTHER SCHEMES

	$o = 1$	$o = 3$	$o = 5$	$o = 8$
Optimization degree on $\mathcal{D}_{T,c}$ VS TRIM	44.33%	92.29%	94.76%	95.55%
Optimization degree on $\mathcal{D}_{T,c}$ VS CMBA-HS	30.75%	93.55%	95.03%	95.40%
Optimization degree on $\mathcal{D}_{T,c}$ VS CMBA-HS++	33.95%	36.11%	41.47%	45.54%

replaces the single-dimensional UCB algorithm with a multi-dimensional UCB algorithm to better adapt to the task environment [25], [30], providing a more refined worker selection strategy.

B. Accuracy of Trust Evaluation

The first experiment compares the trust evaluation accuracy $\mathcal{D}_{\mathcal{T}}$ of MTE-TM, TRIM, CMBA-HS, and CMBA-HS++. $\mathcal{D}_{\mathcal{T}}$ represents the accuracy of workers' expected task data qualification rates. To quantify $\mathcal{D}_{\mathcal{T}}$ on a per-cycle basis, $\mathcal{D}_{T,c}$ is calculated using Eq. 36:

$$\mathcal{D}_{T,c} = \left(\sum_{j=1}^{n_c} \frac{\sum_{k=1}^{o_{j,c}} |p_{j,k} - p_{j,k}^*|}{o_{j,c}} \right) / n_c. \quad (36)$$

Here, n_c and $o_{j,c}$ denote the number of workers who have completed tasks and the number of domains in which worker u_j has completed tasks up to the current cycle, respectively, while $p_{j,k}^*$ represents the true expected value of u_j 's task data qualification rate in domain f_k .

Fig. 4 illustrates the per-cycle evolution of $\mathcal{D}_{T,c}$ for MTE-TM, TRIM, CMBA-HS, and CMBA-HS++ under varying numbers of task domains. The experimental results indicate that MTE-TM outperforms the other three baseline methods. When only one domain is present, all four methods perform reasonably well. However, as the number of domains increases, traditional single-dimensional trust-based mechanisms such as TRIM and CMBA-HS struggle to maintain performance in realistic multi-domain environments. The increasing number of workers and their participation across diverse domains further reduces trust evaluation accuracy. In contrast, MTE-TM and CMBA-HS++ maintain higher accuracy.

Notably, MTE-TM incorporates an iterative trust inference mechanism that updates trust estimates by identifying the workers most likely responsible for the observed outcomes in each task. This mechanism enables MTE-TM to refine trust evaluations over time, resulting in progressively more accurate outcomes as task execution advances.

The experimental results in Table VI demonstrate the maximum improvements achieved by MTE-TM in $\mathcal{D}_{T,c}$ over the other three baseline methods under varying numbers of task domains. Specifically, in the experiments, MTE-TM achieved reductions of 95.55%, 95.40%, and 45.54% compared to TRIM, CMBA-HS, and CMBA-HS++, respectively. These findings further validate the significant advantage of MTE-TM in cross-domain trust evaluation. The comparisons not only highlight the efficiency of the proposed approach but

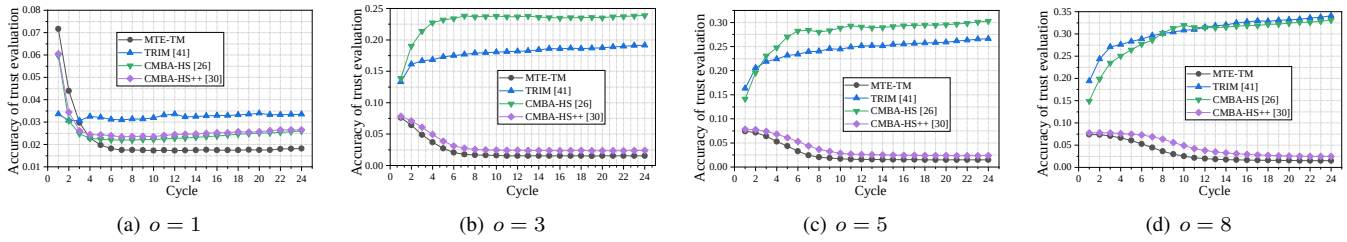


Fig. 4. Accuracy of trust evaluation in different cycles

TABLE VII
DISTRIBUTION OF MALICIOUS/ CREDIBLE WORKERS UNDER DIFFERENT CYCLES BASED ON p_k

Cycle	Proportion	Type	MTE-TM			TRIM			CMBA-HS			CMBA-HS++		
			k=1	k=2	k=3	k=1	k=2	k=3	k=1	k=2	k=3	k=1	k=2	k=3
1	0 ~ 0.1	Malicious	0.56	0.58	0.49	0.31	0.27	0.20	0.26	0.27	0.26	0.57	0.57	0.52
		Credible	0	0	0	0.01	0.01	0.01	0.01	0.01	0.01	0	0	0
	0.9 ~ 1	Malicious	0	0	0	0	0	0	0.01	0.01	0	0	0	0
		Credible	0.23	0.21	0.19	0.53	0.53	0.44	0.33	0.24	0.32	0.20	0.19	0.17
2	0 ~ 0.1	Malicious	0.63	0.63	0.57	0.30	0.25	0.28	0.24	0.24	0.22	0.59	0.62	0.61
		Credible	0	0	0	0.01	0.01	0.01	0.02	0.01	0.02	0	0	0
	0.9 ~ 1	Malicious	0	0	0	0	0	0	0.04	0.01	0.01	0	0	0
		Credible	0.42	0.31	0.29	0.42	0.46	0.40	0.36	0.36	0.37	0.34	0.31	0.20
4	0 ~ 0.1	Malicious	0.85	0.78	0.81	0.29	0.27	0.26	0.16	0.27	0.22	0.80	0.74	0.83
		Credible	0	0	0	0.01	0.01	0.01	0.03	0.02	0.02	0	0	0
	0.9 ~ 1	Malicious	0	0	0	0	0	0	0.03	0.03	0.01	0	0	0
		Credible	0.63	0.55	0.58	0.42	0.42	0.41	0.34	0.34	0.36	0.50	0.48	0.43
8	0 ~ 0.1	Malicious	0.99	0.99	0.98	0.26	0.28	0.29	0.30	0.21	0.31	0.96	0.99	0.97
		Credible	0	0	0	0.01	0.01	0.01	0.03	0.03	0.02	0	0	0
	0.9 ~ 1	Malicious	0	0	0	0	0	0	0	0.01	0.01	0	0	0
		Credible	0.79	0.78	0.80	0.37	0.36	0.35	0.34	0.32	0.32	0.71	0.70	0.70
24	0 ~ 0.1	Malicious	1	1	1	0.29	0.31	0.36	0.32	0.21	0.32	1	1	1
		Credible	0	0	0	0.01	0.02	0.01	0.03	0.03	0.02	0	0	0
	0.9 ~ 1	Malicious	0	0	0	0	0	0	0	0	0.01	0	0	0
		Credible	0.81	0.81	0.80	0.28	0.28	0.28	0.26	0.26	0.26	0.74	0.71	0.72

TABLE VIII
OPTIMIZATION DEGREE OF MTE-TM SCHEME ON $\mathcal{F}_C$ COMPARED WITH OTHER SCHEMES

	$o = 1$	$o = 3$	$o = 5$	$o = 8$
Optimization degree on $\mathcal{F}_C$ VS TRIM	16.44%	88.59%	163.58%	225.09%
Optimization degree on $\mathcal{F}_C$ VS CMBA-HS	4.58%	111.84%	232.60%	351.45%
Optimization degree on $\mathcal{F}_C$ VS CMBA-HS++	6.55%	7.43%	16.55%	18.76%

also underscore the importance of considering multi-domain scenarios when designing incentive mechanisms for spatial crowdsensing systems.

C. F_1 Score

Based on the periodic trust evaluation results of workers, Table VII presents the distribution of trustworthy and malicious workers at the end of each cycle under the condition of three task domains, across four schemes: MTE-TM, TRIM, CMBA-HS, and CMBA-HS++. p_k represents the expected estimated rate of a worker's data in domain k at the end of the current cycle. Each cell in the table shows the proportion of workers

whose p_k falls within two critical intervals: 0~0.1 (malicious) and 0.9~1 (trustworthy), relative to the total number of workers in each category.

The results clearly indicate that MTE-TM outperforms the other three schemes in both the speed and the accuracy of the worker classification. This superiority can be attributed to its “high-penalty, low-reward” trust evaluation mechanism, which modulates the influence of historical performance on the current expectation of u_j through coordination factors $\alpha_{j,k}$ and $\beta_{j,k}$.

For malicious workers, MTE-TM effectively distinguishes them. For trustworthy workers, the evaluation is more rigorous; those who consistently provide high-quality data across multiple tasks can achieve high trust expectations. This design raises the threshold for attaining a high trust value, effectively eliminating borderline workers with unstable performance near the 0.9 threshold, and ensures that only truly reliable workers receive significant weight in the task result aggregation process. Consequently, the overall accuracy and reliability of trust evaluation and decision-making is greatly improved.

To further compare the classification performance of MTE-TM, TRIM, CMBA-HS, and CMBA-HS++, the experiment adopts the F1-score $\mathcal{F}$ to measure the precision-recall balance

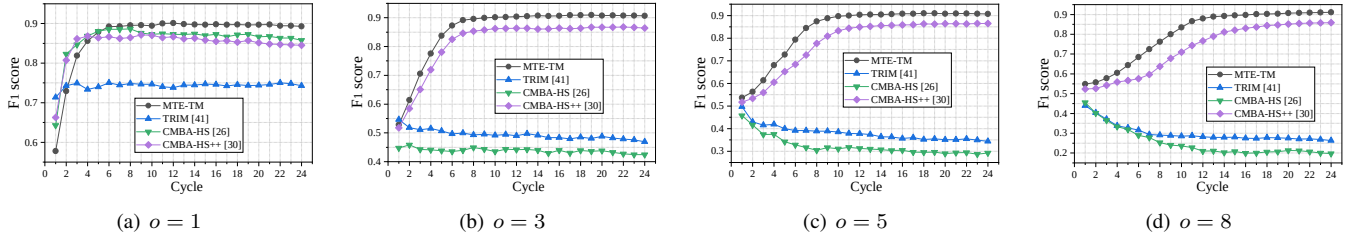


Fig. 5. $\mathcal{F}_1$ score in different cycles

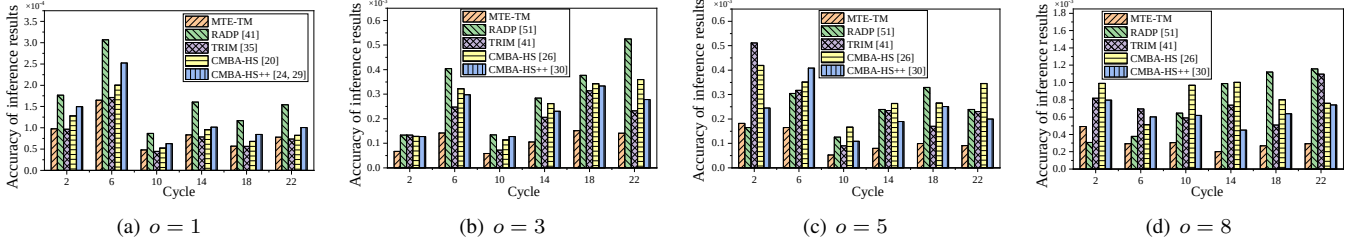


Fig. 6. Accuracy of inference results in different cycles

TABLE IX
OPTIMIZATION DEGREE OF MTE-TM SCHEME ON $\mathcal{D}_{D,c}$ COMPARED WITH OTHER SCHEMES

	$o = 1$	$o = 3$	$o = 5$	$o = 8$
Optimization degree on $\mathcal{D}_{D,c}$ VS RADP	52.91%	71.36%	72.95%	68.26%
Optimization degree on $\mathcal{D}_{D,c}$ VS TRIM	9.78%	60.24%	72.21%	61.90%
Optimization degree on $\mathcal{D}_{D,c}$ VS CMBA-HS	29.46%	53.63%	72.51%	72.44%
Optimization degree on $\mathcal{D}_{D,c}$ VS CMBA-HS++	43.26%	60.50%	60.43%	65.01%

in the classification of workers. To express $\mathcal{F}$ on a per-cycle basis, we use $\mathcal{F}_c$ for the calculation, as shown in Eq. (37):

$$\mathcal{F}_c = \frac{\sum_{c=1}^3 \frac{2 \times \mathcal{P}_c \times \mathcal{R}_c}{\mathcal{P}_c + \mathcal{R}_c}}{3}. \quad (37)$$

Fig. 5 presents the variation of $\mathcal{F}_c$ per cycle for MTE-TM, TRIM, CMBA-HS, and CMBA-HS++ in different domains. The results demonstrate that MTE-TM outperforms the other three baseline methods. As the number of domains and cycles increases, the classification accuracy of traditional single-dimensional trust evaluation schemes such as TRIM and CMBA-HS degrades significantly. In contrast, both MTE-TM and CMBA-HS++ demonstrate better adaptability in multi-domain task environments, with MTE-TM's $\mathcal{F}_c$ achieving higher and faster-converging accuracy across all cycles. Table VIII quantifies the maximum optimization of MTE-TM in $\mathcal{F}_c$ across domains. Specifically, compared to TRIM, CMBA-HS, and CMBA-HS++, MTE-TM improves $\mathcal{F}_c$ by up to 225.09%, 351.45%, and 18.76%, respectively. These results clearly demonstrate MTE-TM's superior performance in balancing classification precision and recall.

D. Accuracy of Inference Results

In crowdsensing networks, the efficiency of task completion is largely determined by the accuracy of the inferred results. To evaluate this, we use the $\mathcal{D}_D$ metric to compare the performance in terms of the accuracy of the inference of five schemes—MTE-TM, RADP, TRIM, CMBA-HS and CMBA-HS++. And we introduce $\mathcal{D}_{D,c}$ as the cycle-normalized representation of $\mathcal{D}_D$, calculated by Eq. (38).

$$\mathcal{D}_{D,c} = \frac{\sum_{i=1}^{m_c} \left(\frac{\sum_{r=1}^{v_i} |D_{i,r} - D_{i,r}^*|}{D_{i,r}^*} \right) / \tau_i}{m_c}. \quad (38)$$

Fig. 6 illustrates the variation of $\mathcal{D}_{D,c}$ after each cycle for the five schemes under different domains. The results demonstrate that MTE-TM outperforms all the comparative schemes. This superior performance is mainly attributed to its iterative solution process that improves the accuracy of the trust evaluation by precisely computing the trust parameters of the worker, thus leading to more accurate inference results. Table IX represents MTE-TM's maximum optimization in $\mathcal{D}_{D,c}$ across domains compared to the other four schemes. Specifically, MTE-TM achieves maximum reductions of 72.95%, 72.21%, 72.51%, and 65.01% in $\mathcal{D}_{D,c}$ relative to RADP, TRIM, CMBA-HS, and CMBA-HS++, respectively. These results clearly demonstrate the superior performance of MTE-TM in improving task inference accuracy and its effectiveness and reliability in handling spatial crowdsensing tasks.

E. Task Budget Saving Rate and Regret

In worker selection, regret is inevitable, since the optimal solution cannot be known in advance. To approximate the optimal outcome while minimizing regret, a balance must be struck between exploration and exploitation. To quantify this, we use $\mathcal{R}_{D,c}$ to measure the performance gap between the MTE-TM and CMBA-HS++ schemes and the theoretical

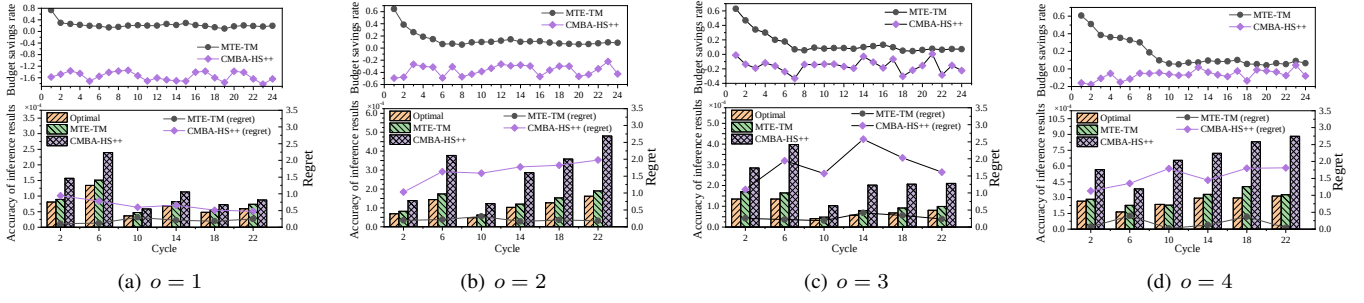


Fig. 7. Variation of $\mathfrak{B}_C$ and $\mathfrak{R}_{D,C}$

TABLE X
OPTIMIZATION DEGREE OF MTE-TM SCHEME ON $\mathfrak{R}_{D,C}$ COMPARED WITH OTHER SCHEMES

	$o = 1$	$o = 3$	$o = 5$	$o = 8$
Optimization degree on $\mathfrak{R}_{D,C}$ VS CMBA-HS++	89.03%	91.26%	88.66%	98.61%

optimal solution. $\mathfrak{R}_{D,C}$ is calculated per cycle as shown in Eq. (39):

$$\mathfrak{R}_{D,C} = \frac{|\mathfrak{D}_{D,C} - \mathfrak{D}_{D,C}^*|}{\mathfrak{D}_{D,C}^*}. \quad (39)$$

Where $\mathfrak{D}_{D,C}^*$ denotes the regret per cycle when using the theoretically optimal set of workers. Additionally, we define $\mathfrak{B}$ to represent the relationship between the task budget and actual payments after task completion, and $\mathfrak{B}_C$ denotes the cycle-normalized version of $\mathfrak{B}$. The formulation is given in Eq. (40):

$$\mathfrak{B}_C = \frac{\sum_{i=1}^{\mathfrak{B}_C} \frac{B_i - \sum_{j=1}^{\mathfrak{B}_C} b_j}{B_i}}{\mathfrak{B}_C}. \quad (40)$$

Fig. 7 illustrates the variations of $\mathfrak{B}_C$ and $\mathfrak{R}_{D,C}$ after each cycle for the MTE-TM and CMBA-HS++ schemes under different domains. The results clearly demonstrate that MTE-TM significantly outperforms CMBA-HS++ in $\mathfrak{R}_{D,C}$. In particular, as the number of domains increases, CMBA-HS++ exhibits notable performance degradation, while MTE-TM maintains consistent performance. This superiority stems from two key factors. First, MTE-TM exhibits significant advantages in inference of task results. Second, unlike CMBA-HS++, MTE-TM employs a more rational approach than simple averaging, enabling precise estimation of task completion probabilities and thereby enhancing the certainty of high-quality task execution. Additionally, due to differences in the generation of this strategy $\mathbb{U}_{\mathbb{F}}$, MTE-TM ensures that actual payments never exceed task budgets after completion. In contrast, CMBA-HS++ relies on worker priorities for selection but fails to adequately balance task budgets and worker expected rewards, which may result in budget overruns. Table X presents the MTE-TM's maximum optimization on $\mathfrak{R}_{D,C}$ over CMBA-HS++ under different numbers of task domains. Specifically, MTE-TM reduces $\mathfrak{R}_{D,C}$ by up to 98.61% compared to CMBA-HS++, clearly demonstrating its superiority in minimizing

regret and improving cost-efficiency. These results strongly validate the effectiveness and cost-efficiency of MTE-TM in task allocation.

VI. CONCLUSION AND FUTURE WORK

A multi-dimensional trust evaluation method is proposed in this paper, extending traditional single-dimensional trust assessment to multiple dimensions. Specifically, workers are evaluated based on multi-dimensional trust metrics for different types of tasks, where each dimension is represented by a probability distribution derived from average performance rather than a single trust score. The Expectation Maximization (EM) algorithm is employed to accurately estimate workers' trust evaluations. The proposed method enables a more precise assessment of workers' performance across different task types, thereby facilitating the selection of workers with the highest quality for specific tasks instead of those with merely high average performance. This provides a solid foundation for prioritizing high-quality workers. Building on this, a new UCB index is designed to reduce unnecessary exploration of task types where a worker performs poorly, thereby lowering platform costs. Finally, an Optimized Data Quality Matching (ODQM) algorithm is proposed to assign tasks to high-quality yet low-bid workers under budget constraints. The experimental results show that the proposed ODQM algorithm outperforms existing methods on four key performance metrics: trust evaluation accuracy $\mathfrak{D}_T$, F1-score $\mathcal{F}$, inference accuracy $\mathfrak{D}_D$, and budget savings rate $\mathfrak{B}$.

Building on the findings of this paper, several promising directions are proposed for future research. First, the development of advanced multidimensional trust models that incorporate temporal and contextual dynamics could further improve worker evaluation accuracy across diverse task types. Second, exploring scalable task allocation methods suitable for larger-scale MCS networks would enhance the practical applicability of the proposed framework. Third, integrating privacy-preserving mechanisms into the multidimensional worker selection process could address growing concerns about data security and user privacy. Finally, extending the ODQM framework to handle heterogeneous data types and complex task requirements can broaden its applicability to more diverse real-world crowdsensing scenarios. These directions provide a solid foundation for advancing high-quality, cost-effective, and reliable task allocation in MCS systems.

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